```c
#include <stdio.h>
#include <string.h>

int main() {
    int data[20], h[30], n, r = 0;
    int i, j, k, pos, error = 0;

    printf("Enter number of data bits: ");
    scanf("%d", &n);

    printf("Enter data bits one by one (MSB first): ");
    for (i = 0; i < n; i++)
        scanf("%1d", &data[i]);

    while ((1 << r) < n + r + 1)
        r++;

    int total = n + r;

    for (i = 1, j = 0; i <= total; i++) {
        if ((i & (i - 1)) == 0)
            h[i] = 0;
        else
            h[i] = data[j++];
    }

    for (i = 1; i <= total; i *= 2) {
        int parity = 0;
        for (j = 1; j <= total; j++) {
            if (j & i)
                parity ^= h[j];
        }
        h[i] = parity;
    }

    printf("Generated Hamming code: ");
    for (i = total; i >= 1; i--)
        printf("%d", h[i]);
    printf("\n");

    printf("Enter received code (%d bits, MSB first): ", total);
    for (i = total; i >= 1; i--)
        scanf("%1d", &h[i]);
```

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    pos = 0;
    for (i = 1; i <= total; i *= 2) {
        int parity = 0;
        for (j = 1; j <= total; j++) {
            if (j & i)
                parity ^= h[j];
        }
        if (parity)
            pos += i;
    }

    if (pos == 0) {
        printf("No error detected.\n");
    } else if (pos <= total) {
        printf("Error detected at bit position %d (from right).\n",
pos);
        h[pos] ^= 1;

        printf("Corrected Hamming code: ");
        for (i = total; i >= 1; i--)
            printf("%d", h[i]);
        printf("\n");
    } else {
        printf("Invalid syndrome; error cannot be corrected.\n");
        error = 1;
    }

    return error;
}
```